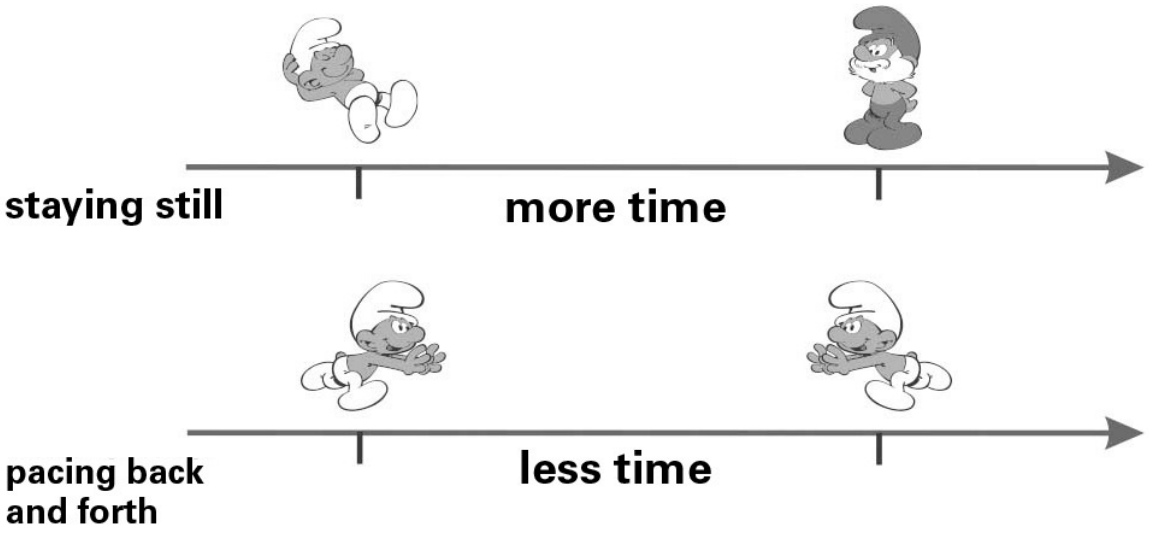


For this effect to become perceptible, one must move very quickly. It was first measured in the 1970s, using precision watches on airplanes.[23] The watch onboard a plane displays a time behind that displayed by the one on the ground. Today, the slowing down of time can be observed in many physics experiments.

In this case, too, Einstein had understood that time slows down *before* the phenomenon was actually measured—when he was just twenty-five years old and studying electromagnetism.

It turned out to be a not particularly complex deduction. Electricity and magnetism are well described by Maxwell's equations. These contain the usual time variable t but have a curious property: if you travel at a certain velocity, then *for you* the equations of Maxwell are no longer true (that is, they don't describe what you measure) *unless* you call "time" a *different* variable: t'.[24] Mathematicians had become aware of this curious feature of Maxwell's equations,[25] but no one had been able to understand what it meant. Einstein grasped its significance: t is the time that passes if I stay still, the rhythm at which things occur that are stationary, like myself; t' is "your time": the rhythm at which things happen that move with you. Thus, t is the time that my watch measures when it is stationary, and t' is the time that your watch measures when it is moving. Nobody had imagined previously that time could be different for a stationary watch and one that was being moved. Einstein had read this within the equations of electromagnetism, by taking them seriously.[26]

A moving object therefore experiences a shorter duration than a stationary one: a watch marks fewer seconds, a plant grows more slowly, a young man dreams less. For a moving object, time contracts.* Not only is there no single